

AI Friday Season 2 – Problem Statements

AI-Powered Market Sentiment Summarizer for Capital Markets

PROBLEM STATEMENT

Traders and analysts in capital markets often face an overwhelming amount of daily news, reports, and social media content, making it difficult to quickly grasp market sentiment. This information overload delays decision-making and can lead to missed opportunities or increased risk exposure. Manual sentiment analysis is time-consuming and inconsistent across different data sources. There is a need for an automated system that can rapidly synthesize and summarize market sentiment from diverse textual sources. The solution should help users prioritize information and react faster to market changes.

Data Considerations (Guidelines):

Financial news headlines and article summaries from public RSS feeds (e.g., Reuters, Bloomberg), social media posts filtered by finance-related hashtags or stock tickers (e.g., Twitter API), historical sentiment labels from public datasets for validation, and sector/ticker metadata from open financial databases.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

A GenAI agent that processes aggregated textual data daily to produce sector-wise and ticker-specific sentiment summaries via a simple web UI. Deliverables include text summaries highlighting positive, neutral, and negative sentiment trends, a dashboard displaying sentiment scores, and a demo video showcasing the agent's ability to answer user queries about market sentiment. Success measured by user testing feedback on summary relevance and time saved in sentiment analysis.

Be creative to go beyond as you solve the above needs

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Performance Bottleneck Explanation Bot for Application Engineering

PROBLEM STATEMENT

Performance engineers often receive complex logs and metrics from applications experiencing slowdowns but lack immediate insights into root causes. Analyzing these performance issues manually requires expert knowledge and is time-consuming, delaying resolution and impacting user experience. There is a need for an AI assistant that can interpret performance data and provide understandable explanations and remediation steps quickly.

Data Considerations (Guidelines):

Sample performance logs (CPU, memory usage, response times), trace files, configuration metadata from open-source applications, or generated synthetic data simulating bottlenecks. Data can be loaded from files or APIs.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

An AI assistant generating detailed textual reports explaining detected performance issues and prioritized recommendations, accessible via a simple web interface. Deliverables include generated reports, UI screenshots, and a demo video. Success measured by reduction in diagnosis time and positive feedback from performance engineers.

Be creative to go beyond as you solve the above needs

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AI-Driven Insurance Claim Explanation Generator

PROBLEM STATEMENT

Insurance customers often find claim decisions confusing due to complex policy language and technical jargon. Customer service teams spend significant time explaining claim outcomes, leading to increased operational costs and customer dissatisfaction. There is a need for an AI assistant that can generate clear, personalized explanations of claim decisions to improve transparency and reduce support workload.

Data Considerations (Guidelines):

Sample anonymized claim records, policy documents in text format, common customer queries and FAQs. Data can be sourced from public datasets or simulated for privacy reasons.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

A GenAI-powered assistant producing easy-to-understand textual explanations of claim decisions tailored to individual customers, accessible via a simple UI. Deliverables include generated explanation texts, usage documentation, and a demo video. Success measured by customer comprehension scores and reduced support interaction times.

Be creative to go beyond as you solve the above needs

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Life Sciences Research Abstract Simplifier

PROBLEM STATEMENT

Researchers and clinicians often face difficulty quickly comprehending complex scientific abstracts due to dense terminology and jargon. This slows knowledge dissemination and collaborative decision-making. There is a need for an AI tool that can generate simplified, layman-friendly summaries of research abstracts to accelerate understanding across diverse stakeholders.

Data Considerations (Guidelines):

Open-access life sciences research abstracts from repositories like PubMed, domain-specific glossaries, and sample simplified summaries for reference. Data can be accessed via APIs or bulk downloads.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

An AI agent that takes scientific abstracts as input and outputs concise, plain language summaries via a simple interface. Deliverables include example summaries, user documentation, and a demo video. Success measured by user comprehension scores and time saved in reviewing literature.

Be creative to go beyond as you solve the above needs

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Automated Application Bug Report Summarizer for Maintenance Teams

PROBLEM STATEMENT

Application maintenance teams receive numerous verbose bug reports that are difficult to triage quickly. Key information is often buried in lengthy descriptions, slowing issue prioritization and resolution. There is a need for an AI assistant to extract and summarize critical bug details to streamline maintenance workflows.

Data Considerations (Guidelines):

Sample bug reports from open issue trackers (e.g., GitHub, Jira), metadata on application modules, and severity classification labels. Data can be anonymized and pre-processed.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

An AI agent producing structured textual summaries of bug reports accessible through a minimal interface. Deliverables include summary samples, documentation, and a demo video. Success measured by triage time reduction and summary accuracy ratings.

Be creative to go beyond as you solve the above needs

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Data Analytics Insight Generator from KPI Dashboards

PROBLEM STATEMENT

Business analysts spend time interpreting KPI dashboards to derive insights, which can be subjective and inconsistent. Automating the generation of natural language insights from dashboard data can standardize reporting and speed decision-making. Existing tools often require manual input or complex configuration.

Data Considerations (Guidelines):

Sample KPI datasets in JSON or CSV format, historical trends, and domain-specific threshold values. Data can be simulated or publicly sourced.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

An AI agent generating textual analytical insights from uploaded KPI data accessible via a simple interface. Deliverables include generated insight reports, usage instructions, and a demo video. Success measured by user feedback and reduced analysis time.

Be creative to go beyond as you solve the above needs



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Back Office Document Summarizer for Financial Operations

PROBLEM STATEMENT

Back office teams in financial institutions handle large volumes of documents such as contracts, transaction records, and regulatory filings. Manually reviewing these documents to extract key information is time-consuming and error-prone. Automating document summarization can improve operational efficiency and reduce risk of oversight.

Data Considerations (Guidelines):

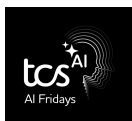
Sample financial contracts, transaction documents in PDF or text format, and annotated key information datasets. Data can be sourced from public domain or simulated.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

An AI-powered summarizer producing concise textual highlights of uploaded documents accessible through a simple interface. Deliverables include summary samples, usage documentation, and a demo video. Success measured by accuracy and time savings.

Be creative to go beyond as you solve the above needs



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Capital Markets Regulatory Disclosure Summarizer

PROBLEM STATEMENT

Regulatory disclosures in capital markets are lengthy and complex, making it difficult for compliance officers to quickly identify key changes and implications. Manual review is time-consuming and prone to oversight. Automated summarization can improve compliance and reduce risk.

Data Considerations (Guidelines):

Sample regulatory filings, disclosures, and compliance guidelines in text format. Data can be sourced from public regulatory agency websites.

Feel free to generate synthetic data where appropriate using the provided environment

SOLUTION EXPECTATIONS

An AI-powered summarizer producing concise, actionable summaries of regulatory disclosures accessible via a simple UI. Deliverables include summary reports, usage documentation, and a demo video. Success measured by accuracy and review speed improvement.

Be creative to go beyond as you solve the above needs
